r_{jm} times, such that the total (over all j) number of times each part m occurs is i_m. Thus we get

$$\chi_{U_\lambda}(C_{\mathbf{i}}) = \sum_r \prod_m \frac{i_m!}{\prod_j r_{jm}!}.$$

But this is exactly the coefficient of x^λ in

$$\prod_{m\geq 1} (x_1^m + \cdots + x_N^m)^{i_m}$$

(r_{jm} is the number of times we take x_j^m). □

5.15. The Frobenius character formula

Let $\Delta(x) = \prod_{1\leq i<j\leq N}(x_i - x_j)$. Recall that $\Delta(x)$ is the Vandermonde determinant, $\det(x_i^{N-j})$. Let $\rho = (N-1, N-2, \ldots, 0) \in \mathbb{C}^N$. The following theorem, due to Frobenius, gives a character formula for the Specht modules V_λ.

Theorem 5.15.1. *Let $N \geq p$. Then $\chi_{V_\lambda}(C_{\mathbf{i}})$ is the coefficient of $x^{\lambda+\rho} := \prod x_j^{\lambda_j+N-j}$ in the polynomial*

$$\Delta(x) \prod_{m\geq 1} H_m(x)^{i_m}.$$

Remark 5.15.2. Here is an equivalent formulation of Theorem 5.15.1: $\chi_{V_\lambda}(C_{\mathbf{i}})$ is the coefficient of x^λ in the (Laurent) polynomial

$$\prod_{i<j}\left(1 - \frac{x_j}{x_i}\right) \prod_{m\geq 1} H_m(x)^{i_m}.$$

Proof. For brevity denote χ_{V_λ} by χ_λ. Let us denote the class function defined in the theorem by θ_λ. We claim that this function has the property $\theta_\lambda = \sum_{\mu\geq\lambda} L_{\mu\lambda}\chi_\mu$, where $L_{\mu\lambda}$ are integers and $L_{\lambda\lambda} = 1$. Indeed, from Theorem 5.14.3 we have

$$\theta_\lambda = \sum_{\sigma\in S_N} (-1)^\sigma \chi_{U_{\lambda+\rho-\sigma(\rho)}},$$

where if the vector $\lambda+\rho-\sigma(\rho)$ has a negative entry, the corresponding term is dropped, and if it has nonnegative entries which fail to be nonincreasing, then the entries should be reordered in nonincreasing order, making a partition that we'll denote by $\langle\lambda+\rho-\sigma(\rho)\rangle$ (i.e., we